

YANHUI YANG

yyang440@ucr.edu $\diamond$ <https://astro-yyh.github.io>

EDUCATION

PhD Program in Physics University of California, Riverside
Concentration in Astronomy
Overall GPA: 4.00/4.00

Sep. 2022–present

BS in Astronomy University of Science and Technology of China
Wang Shouguan Talent Program in Astronomy
Thesis: *Hydrodynamic simulations of turbulent mixing layers of galactic winds*
Overall GPA: 3.74/4.30, Major GPA: 3.93/4.30 (Physics & Astronomy courses)

Jul. 2021

RESEARCH INTERESTS

Cosmology: large-scale structure, dark matter, dark energy, cosmological emulation; Astrophysics: galaxy formation and evolution, circumgalactic medium, turbulent mixing layers

APPOINTMENTS

Graduate Student Researcher, Advisor: Simeon Bird (UCR)

Fall 2023–present

Teaching Assistant for *General Physics Laboratory 2LC*, Coordinator: Jonathan Eldridge (UCR)
2023

Winter–Spring

Research Assistant, Advisor: Suoqing Ji (SHAO)

Spring 2022

PUBLICATIONS

Yanhui Yang, Simeon Bird, Yihao Zhou, Tiziana Di Matteo, Rupert Croft, Yueying Ni, and Nianyi Chen. Matter Clustering in Astrid: Reduced Baryonic Suppression from Realistic Black Hole Dynamics. *ApJL*, 1007(1):L20, August 2026. doi: 10.3847/2041-8213/ae8dc6.

Yanhui Yang, Simeon Bird, Ming-Feng Ho, and Mahdi Qezlou. Ten-dimensional neural network emulator for the nonlinear matter power spectrum. *Phys. Rev. Lett.*, 136(6):061001, February 2026a. doi: 10.1103/8yzy-t59w.

Yanhui Yang, Simeon Bird, Ming-Feng Ho, and Mahdi Qezlou. Design and optimization of neural networks for multifidelity cosmological emulation. *Phys. Rev. D*, 113(4):043508, February 2026b. doi: 10.1103/cqrc-k8wq.

Yanhui Yang, Simeon Bird, and Ming-Feng Ho. Ten-parameter simulation suite for cosmological emulation beyond Λ CDM. *Phys. Rev. D*, 111(8):083529, April 2025. doi: 10.1103/PhysRevD.111.083529.

Yanhui Yang and Suoqing Ji. Radiative turbulent mixing layers at high Mach numbers. *MNRAS*, 520(2):2148–2162, April 2023. doi: 10.1093/mnras/stad264.

Only first-author papers are listed above. Please refer to [NASA/ADS](#) (ORCID: 0000-0001-6221-6024) for a complete list of my publications.

PRESENTATIONS

Invited Talks

Weak Baryonic Suppression in Astrid: A Challenge for Precision Cosmology

University of Kansas Astronomy & Space Physics Seminar, The University of Kansas

Sep. 2026

Modeling Cosmic Large-Scale Structure Beyond Λ CDM with Simulations and Emulation

Fudan-CfAA Seminar, Fudan University

Oct. 2025

<i>Goku: A 10-parameter simulation suite for cosmological emulation</i> Physics and Astronomy Student Seminar, University of California, Riverside	Mar. 2025
<i>Radiative turbulent mixing layers at high Mach numbers</i> Shanghai Astronomical Observatory, Chinese Academy of Sciences	May 2022

Contributed Talks and Posters

<i>Weak Baryonic Suppression in Astrid: A Challenge for Precision Cosmology</i> Cosmology from Home 2026	Sep. 2026
<i>Cosmological Emulation based on the Goku Simulations in a 10-Dimensional Parameter Space</i> XIX International Conference on Interconnections between Particle Physics and Cosmology (PPC2026) The University of New South Wales, Sydney	Aug. 2026
<i>Weak Baryonic Suppression in Astrid: A Challenge for Roman Weak Lensing Cosmology</i> Shifting Landscapes in Astrophysics: New Frontiers to Explore with Roman California Institute of Technology/IPAC	Jul. 2026
<i>Improving Cosmological Emulation via Multifidelity Neural Networks</i> Galaxy Formation & Evolution in Southern California, University of California, San Diego	Sep. 2025
<i>Nonlinear Matter Power Spectrum Emulation beyond ΛCDM for Next-Generation Surveys (poster)</i> Cosmic Cartography with Roman 2025, STScI, Baltimore, MD	Jul. 2025
<i>10D emulation for the nonlinear matter power spectrum beyond ΛCDM based on Goku</i> Cosmology from Home 2025	Jul. 2025
<i>Goku: A 10-parameter simulation suite for cosmological emulation</i> Frontera User Meeting 2024, TACC, The University of Texas at Austin	Aug. 2024

ADVISORS & REFERENCES

Simeon Bird (sbird@ucr.edu) — PhD advisor Associate Professor, Department of Physics and Astronomy, University of California, Riverside
Tiziana Di Matteo (tiziana@phys.cmu.edu) Professor, Department of Physics, Carnegie Mellon University
Rupert A. C. Croft (rcroft@cmu.edu) Professor, Department of Physics, Carnegie Mellon University
Suoqing Ji (sqji@fudan.edu.cn) Associate Professor, Department of Physics, Fudan University
Philip F. Hopkins (phopkins@caltech.edu) Professor, Division of Physics, Mathematics and Astronomy, California Institute of Technology
Xiaobo Dong (xbdong@ynao.ac.cn) Research Professor, Yunnan Observatories, Chinese Academy of Sciences

AWARDS & HONORS

Dean's Distinguished Fellowship (University of California, Riverside)	2022
NAOC Scholarship (National Astronomical Observatories, Chinese Academy of Sciences)	2020
National Encouragement Scholarship	2020
Outstanding Student Scholarship (Grade 3)	2019

MENTORSHIP & SUPERVISION

Undergraduate students (co-mentored with Prof. Simeon Bird): Mark Achenbach (UH Mānoa), Michael Padilla (UCR), Mohak Bhattacharya (UCR)

PROFESSIONAL SERVICE

Journal referee for *Physical Review D* and *Physical Review Letters*

SKILLS

Programming languages	C (GSL), C++ (Boost, CUDA), Python (PyTorch, mpi4py, PyMC, pandas, NumPy, Matplotlib, yt, BigFile, GPy, etc.), Bash, Fortran, HTML, CSS, Markdown, $\text{\LaTeX}$
Simulation codes	MP-Gadget, CLASS, FLASH, GADGET-3, RAMSES, AREPO
Tools	Git, Mathematica, Origin, Gnuplot, IDL, DS9, VS Code

VISITS & ADDITIONAL TRAINING

TACC Open Hackathon, Texas Advanced Computing Center, UT Austin	Oct. 2024
The 2023 Machine Learning Institute, TACC, UT Austin	Jun. 2023
Shanghai Astronomical Observatory, Chinese Academy of Sciences	Jan. 2020
Five-hundred-meter Aperture Spherical Telescope (FAST)	Jul. 2019

OUTREACH

Physics Fun Expedition: A Science Outreach Program for Township Middle Schools	Jul. 2019
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